

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)

MARKET RISK MANAGEMENT AND REGULATION

(RISK MEASUREMENT, PROPERTIES OF RISK MEASURES, RISK MEASUREMENT OF STOCKS, BONDS, DERIVATIVES AND PORTFOLIOS, CAPITAL REQUIREMENTS FOR MARKET RISK, FRTB)

Sources:

- John C. Hull - Risk Management and Financial Institutions (2015): Chapter 12, 13 and 14
- Some explanations from Investopedia, CFA Institute websites

Definition: Market risk is the risk of losses on financial investments caused by adverse price movements. Examples of market risk are: changes in equity prices, commodity prices, interest rate moves or foreign exchange fluctuations.

Greeks

Front office hedges risks by ensuring that exposures to individual market variables are not too high. Each of the greeks measures a different aspect of the risk in a trading book.

Delta (Δ) shows the sensitivity of a portfolio to the price of the underlying:

$$\text{Delta} = \frac{dP}{dS}$$

A *linear product* is one whose value at any given time is linearly dependent on the value of an underlying market variable, ex. spot position in the asset, forward contract. For such products, hedges protect against large changes as well as small ones in the value of the underlying (“hedge and forget”). For *non-linear products*, ex. options, achieving delta neutrality only protects against small movements in the price of the underlying asset (dynamic hedging, rebalancing). When the stock price is low, delta of an option is close to zero, when it is high, the probability of exercising is higher, so it’s close to 1.

Gamma (Γ) of a portfolio measures the rate of change of the portfolio’s delta with respect to the price of the underlying (how frequently is rebalancing needed):

$$\text{Gamma} = \frac{d^2 P}{d^2 S}$$

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)

If gamma is small, delta changes slowly, if gamma is large, delta is highly sensitive to the price of the underlying. It is positive for a long position in an option.

A *linear product* has zero gamma. To make a portfolio gamma neutral, we need non-linear products, like a traded option with position of $-\Gamma / \Gamma_T$. Keeping gamma neutrality also requires frequent rebalancing.

Vega (V) is the rate of change of the value of the portfolio with respect to the volatility of the underlying asset price.

$$\text{Vega} = \frac{dP}{d\sigma}$$

If vega is high, the portfolio is very sensitive to small changes in volatility. The vega of a long position in an option is positive.

Vega can be changed by adding a position of $-V / V_T$ in a traded option, thus reaching vega neutral position. A portfolio which is gamma neutral won't be vega neutral and vice versa with one option but using two options, we can reach both of them.

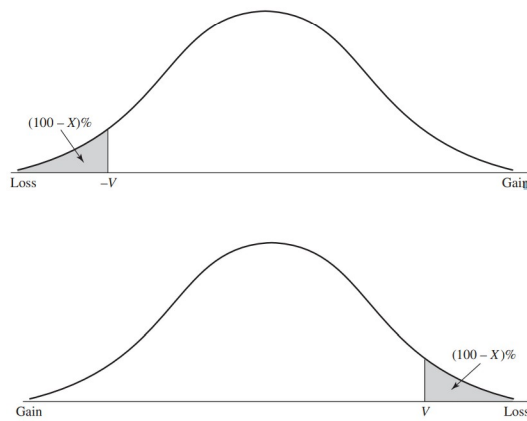
VAR and ES

Often a financial institution's portfolio depends on hundreds, or even thousands, of *market variables* (e.g., an equity index, an interest rate, or a commodity price). A huge number of risk measures is therefore produced each day, such as delta, gamma, or vega. While very useful to traders, the risk measures are not useful to top management. Value at risk (VaR) and Expected Shortfall (ES) are attempts to provide a single number that summarizes the total risk in a portfolio.

VaR is the *loss level that will not be exceeded during a time period of length T with an $X\%$ CI under normal circumstances*. It asks the question: “**How bad can things get?**”. VaR of a portfolio is a function of two parameters: the *time horizon*, T , and the *confidence level*, $X\%$. VaR can be calculated from (either the probability distribution of gains during time T or) the probability distribution of losses during time T . (In the former case, losses are negative gains; in the latter case, gains are negative losses.)

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)



$$\text{VaR} = \mu + \sigma N^{-1}(X)$$

For relatively short time horizons, μ is often assumed to be zero. VaR for a particular confidence level is then proportional to σ .

Advantages of VaR

- It captures an important aspect of risk in a single number
- Easy to understand and interpret

Drawback of VaR

- When VaR is used to limit the risks taken by a trader, it can lead to unacceptable risk.
- Many trading strategies give high probability of good returns and small probability of a huge loss, ex. huge, short positions in OTM options.

ES is the *expected loss / average loss during time T conditional on the loss being greater than the X th percentile of the loss distribution*. ES asks: **“If things do get bad, what is the expected loss?”**. Setting an ES rather than a VaR limit for traders makes it less likely that they will be able to take unacceptable risks.

When the loss is assumed to be normally distributed with mean μ and standard deviation σ , ES with a confidence level of X is given by

$$ES = \mu + \sigma \frac{e^{-Y^2/2}}{\sqrt{2\pi}(1 - X)}$$

This shows that when μ is assumed to be zero, ES is proportional to σ .

Drawback of ES

- It does not have the simplicity of VaR and is more difficult to understand
- More difficult to back-test

Back-testing

It is a test of *how well the current procedure for calculating the risk measure would have worked in the past*. VaR is easier to back-test than ES. Back-test involves looking at a period in the past that could have a huge negative impact on our portfolio. Then check how often the loss in a day (for a 1-day VaR) exceeded the VaR. Such days are *exceptions*. If exceptions happen only a few times, it's okay. If they happen frequently, it means the VaR is *underestimated*.

Properties of a risk measure

- **Monotonicity**: if a portfolio produces a worse result than another portfolio for every state of the world, its risk measure should be greater.
- **Translation invariance**: If an amount of cash K is added to a portfolio, its risk measure should decrease by K .
- **Homogeneity**: Changing the size of a portfolio by a factor X keeping the relative asset weights the same should lead to a risk measure multiplied by X .
- **Subadditivity**: The risk measure for two portfolios after they have been merged should be no greater than the sum of their risk measures separately.

VaR satisfies the first 3 conditions. Those which satisfy all four are *coherent risk measures*. ES is always coherent.

1. Value at Risk (VaR):

- **Monotonicity**: VaR satisfies monotonicity. If portfolio A has worse outcomes than portfolio B , then

$$\text{VaR}(A) \geq \text{VaR}(B).$$

- **Translation Invariance**: VaR satisfies translation invariance.
- **Positive Homogeneity**: VaR satisfies positive homogeneity.
- **Sub-additivity**: VaR does **not** always satisfy sub-additivity. For some distributions, VaR can be super-additive, meaning diversification might not reduce risk. This is a key reason VaR is often criticized.

Conclusion: VaR is **not** a coherent risk measure because it does not satisfy sub-additivity in all cases.

2. Expected Shortfall (ES) (also known as Conditional VaR or CVaR):

- **Monotonicity:** ES satisfies monotonicity.
- **Translation Invariance:** ES satisfies translation invariance.
- **Homogeneity:** ES satisfies positive homogeneity.
- **Sub-additivity:** ES satisfies sub-additivity, as it accounts for extreme tail risks and is always sub-additive.

Conclusion: ES is a **coherent** risk measure because it satisfies all four properties.

3. Variance:

- **Monotonicity:** Variance does **not** satisfy monotonicity. A portfolio with a high variance does not necessarily imply worse outcomes than a portfolio with low variance, as variance measures the spread of returns, not their downside risk.
- **Translation Invariance:** Variance does **not** satisfy translation invariance because adding a constant to a portfolio does not reduce its variance.
- **Homogeneity:** Variance does **not** satisfy positive homogeneity. If you scale a portfolio by λ , the variance scales by λ^2 , not λ .
- **Sub-additivity:** Variance satisfies sub-additivity for independent risks, but not in all cases, particularly when risks are correlated in specific ways.

Conclusion: Variance is **not** a coherent risk measure because it fails monotonicity, homogeneity, and translation invariance.

4. Standard Deviation:

- **Monotonicity:** Standard deviation does **not** satisfy monotonicity. This is because it measures the spread of returns around the mean, not just downside risk. A portfolio with high standard deviation doesn't necessarily imply worse outcomes than one with low standard deviation, so it fails to reflect the downside risk accurately.
- **Translation Invariance:** Standard deviation does **not** satisfy translation invariance because adding a constant to all outcomes of a portfolio does not change the variability (spread) around the mean. Therefore, adding a risk-free amount to a portfolio does not reduce its standard deviation.
- **Homogeneity:** Standard deviation **satisfies** homogeneity, meaning that scaling a portfolio by a factor $\lambda > 0$ scales the standard deviation by λ . So this property is met.

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)

- **Sub-additivity:** Standard deviation satisfies sub-additivity for independent or certain types of correlated risks but **not universally**. For some combinations of risks, standard deviation may not be sub-additive. This limitation can make it unreliable in representing risk reduction due to diversification, particularly when dependencies between assets aren't well-behaved.

Conclusion: Standard deviation is **not** a coherent risk measure, as it fails to meet the criteria of monotonicity, sub-additivity in all cases, and translation invariance.

Parameters

An appropriate choice for the **time horizon T**, when VaR or ES is calculated, depends on the application. When positions are very liquid and actively traded, it makes sense to use a short time horizon (perhaps only a few days). In such cases μ is assumed to be zero. When market risk is considered, analysts often start with a 1-day VaR and ES, then scale them. Longer time horizons may not be meaningful because of changes in the composition of the portfolio. Anyway, as the time horizon increases, the VaR increases too.

$$T\text{-day VaR} = 1\text{-day VaR} \times \sqrt{T}$$
$$T\text{-day ES} = 1\text{-day ES} \times \sqrt{T}$$

These formulas are exactly true when the changes in the value of the portfolio on successive days have *independent identical normal distributions* with mean zero. In other cases, they are approximations.

The choice of **CI** depends on a number of factors. A company could check the PD for a certain period of time according to its credit rating and calculate VaR at a CI of 1-PD to see how much it has to lose to become insolvent. In the end it is a choice based on risk attitude. Anyway, as CI increases, VaR increases too. The capital requirement for market risk has been based on a **10-day 99% CI VaR**.

Marginal VaR

The *marginal VaR* for a subportfolio is the sensitivity of VaR to the amount invested in this subportfolio. We can calculate it by increasing the invested amount in the portfolio by a small amount, and then recalculate the VaR to see how it changed. For a well-diversified investment

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)

portfolio, *marginal VaR is closely related to the CAPM's beta*. If an asset's beta is high, its marginal VaR will tend to be high. If its beta is low, the marginal VaR tends to be low.

Incremental VaR

The *incremental VaR* is the difference between VaR with the subportfolio and VaR without the subportfolio.

Aggregating VaRs and ESs

$$\text{VaR}_{\text{total}} = \sqrt{\sum_i \sum_j \text{VaR}_i \text{VaR}_j \rho_{ij}}$$

where VaR_i is the VaR for the i th segment, $\text{VaR}_{\text{total}}$ is the total VaR, and ρ_{ij} is the correlation between losses from segment i and segment j . This is exactly true when the losses (gains) have zero-mean normal distributions and provides a good approximation in many other situations. The same is true when VaR is replaced by ES in equation.

Spectral Risk Measures

- *Assigns weights to quantiles of the loss distribution*
- VaR assigns 100% weight to the X th percentile of the loss distribution
- ES assigns equal weight to all percentiles greater than the X th percentile of the loss distribution
- Are coherent if weights are non-decreasing

Risk Measurement of...

VaR of stocks

A procedure called *mapping* could be used to calculate the VaR. This consists of decomposing a financial instrument or portfolio into *risk factors*. We can map a stock market index into the individual stock composing the index. However, in practice, they are not mapped but used as risk factors.

For a portfolio with **N identical shares** we can use the *delta-normal method* to calculate its VaR. The risk factor of a share is its *return*, which is assumed to be normally distributed. If short $\mu=0$, then

$$\text{VaR} = \text{Multiplier} * \text{Position} \cdot \cdot$$

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios,
capital requirements for market risk, FRTB)

where $Multiplier = N^{-1}(1-CI)$.

For a portfolio with **n different shares** we can use the *delta-normal method* to calculate its VaR. We calculate the weights for each risk factor (share) in the portfolio, the returns assumed to be normally distributed, so the portfolio returns are also normally distributed.

$$VaR = Multiplier * \sigma_v = Multiplier * \sqrt{wCw} = Multiplier * \sqrt{(w\sigma) R (w\sigma)}$$

VaR of bonds

A procedure called *mapping* could be used to calculate the VaR for bonds / bond portfolio. This consists of decomposing a financial instrument or portfolio into risk factors. In case of bonds the risk factors are *interest rates*, so we want to know how bond prices change in response to interest rate movements (duration).

There are 3 approaches:

- **Duration approach**

- Mapping to a *single interest rate* (YTM or $y = r$)
- We assume linear relationship between dP and dy , assuming parallel shifts; it neglects convexity! We also assume that daily returns are normal and IID.

$\Delta P = -D_{\text{mod}} \cdot P \cdot \Delta r + \left(\frac{1}{2} \cdot Cx \cdot P \cdot \Delta r^2 + \circ \right)$, where

$D_{\text{mod}} = \frac{D}{(1 + YTM)}$
P: bond price

Delta- $VaR_{\text{bond}} = D_{\text{mod}} * P * VaR(\%)_{\text{yield}}$

- $VaR(\%)_{\text{yield}}$ is the quantile-specific worst daily movement in yield-returns

$$\text{worst yield}_{\text{return}} = \sigma * z_{\text{statistic}}$$

- After calculating the duration of each bond, define various scenarios for potential interest rate changes over a chosen time horizon
- Calculate the corresponding change in value for each bond. By summing these changes, you estimate the overall impact on the entire portfolio's value.
- Rank the potential portfolio losses from these scenarios and identify the loss at your chosen confidence level (e.g., 95%), that will be the VaR

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)

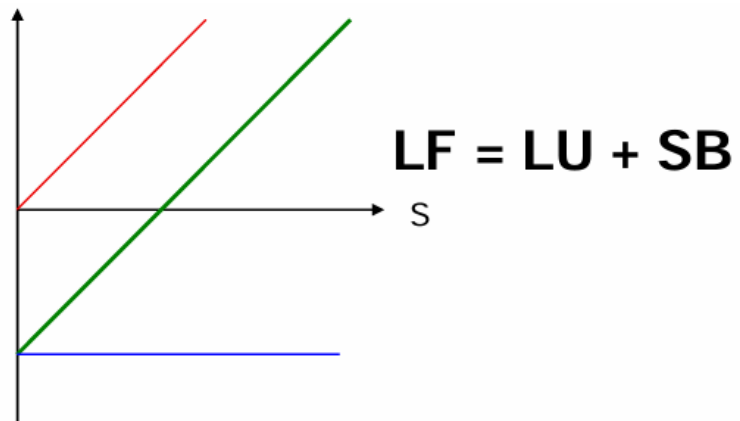
- *Duration approach uses duration as a proxy for a bond's interest rate risk and estimates the portfolio's VaR based on potential interest rate changes*
- **Cash flow mapping**
 - Mapping to a *series of interest rates* ($r_1, r_2, \dots$ = yield curve)
 - Market variables are *zero-coupon bond prices* with different maturities
 - Instead of using duration, this method focuses on the individual cash flows of each bond in the portfolio. We strip the bonds into separate CFs (stripped zero-coupon bonds)
 - We choose zero bonds with standard maturities, and we calculate the needed interest rates by interpolation to get the PV of CFs
 - After that we can calculate the avg return and SD of the portfolio to get the VaR
 - *Cash Flow Mapping decomposes the portfolio into individual cash flows, maps them to zero-coupon bonds, and simulates market movements to estimate the VaR with greater precision*
- **PCA**
 - It helps reduce the complexity of the yield curve consisting of multiple maturities into a few main uncorrelated components that capture the majority of the variance.
 - The first PCA is the overall level (roughly parallel movement) of the yield curve, the second is the change in the slope and the third is the curvature
 - The change in the portfolio value for 1bp movements in the different rates is expressed in dollars. Then the exposure to the principal components can be calculated per bp. This can be used to calculate the SD of the factor score leading to the result of the VaR calculation.
 - *PCA uses historical return data to identify underlying risk factors and estimate VaR by simulating changes in these factors.*

VaR of forwards

Mapping into underlying asset and interest rate position.

MARKET RISK MANAGEMENT AND REGULATION

*(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios,
capital requirements for market risk, FRTB)*



MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)

VaR of options

We can map to *full evaluation* where we can use the BS model and can estimate VaR by Monte Carlo simulation. We value the portfolio today and sample once from the multivariate distribution of the Δx_i . We use these to determine the market variables at the end of the day so we can calculate the portfolio value EOD. Afterwards we calculate ΔP , and repeat the procedure many times to have a probability distribution for ΔP . Then we can find the VaR which corresponds to the chosen CI.

Alternatively, we can map to *Greek letters* when delta normal VaR can be calculated (using delta and gamma): Δ

$$VaR(dc) = 1,645 \cdot \sqrt{\Delta^2 \cdot S^2 \cdot \sigma^2 + \frac{1}{2} \Gamma \cdot S^2 \cdot \sigma^2}$$

Delta (Δ) of a portfolio is the partial derivative of a portfolio with respect to the price of the underlying asset. *Gamma* (Γ) is the rate of change of delta with respect to the price of the underlying asset. It measures the delta hedging errors caused by curvature.

VaR and ES estimation

Historical Simulation Approach

Being a non-parametric method, it is the most popular approach to calculate VaR and ES for market risk. Historical simulation *uses past data* to see what could happen in the future and these past data provide the distribution. Given equal probability for each outcome, the quantile defined is the VaR. Exponentially decreasing weights could also be used. We *do not make any assumptions about the distribution*, but we assume that *things can only happen in the future as they did in the past*.

The first step is to *identify the market variables* affecting the portfolio: exchange rates, interest rates, stock indices, and so on. *Data are then collected on movements in these market variables* over the most recent 501 days. This provides 500 *alternative scenarios* for what can happen between today and tomorrow. For each scenario, the *dollar change in the value* of the portfolio between today and tomorrow is calculated. This defines a *probability distribution for daily loss* (with gains counted as negative losses) in the value of the portfolio. Given equal probability for each outcome (we could weigh the observations, thus having exponentially decreasing weight

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)

as the data become older), the 99 percentile of the distribution can be estimated as the fifth worst outcome. To calculate ES using historical simulation, we *average the observations* in the tail of the distribution of losses.

Advantage:

- historical data determine the joint probability distribution of the market variables (*no assumption about the distribution*, we just assume that future will be like past)

Disadvantage:

- computationally much slower than the model-building approach

Stressed VaR and ES can be calculated from historical data. The calculations given so far assumed that the most recent data is used for the historical simulation on any given day. However, historical simulations can be based on data from any period in the past. Periods of high volatility will tend to give high values for VaR and ES, whereas periods of low volatility will tend to give low values.

The *standard error* for a VaR that is estimated using historical simulation tends to be quite high. However, as the sample gets larger, the error will decrease. The higher the VaR confidence level required, the higher the standard error. In one extension of the basic historical simulation approach, the weights given to observations decrease exponentially as the observations become older; in another, adjustments are made to historical data to reflect changes in volatility.

Extreme value theory is a way of smoothing the tails of the probability distribution of portfolio daily changes calculated using historical simulation. It leads to estimates of VaR and ES that reflect the whole shape of the tail of the distribution, not just the positions of a few losses in the tails. Extreme value theory can also be used to estimate VaR and ES when the confidence level is very high. For example, even if we have only 500 days of data, it could be used to come up with an estimate of VaR or ES for a confidence level of 99.9%.

Model-Building Approach / Parametric method / Analytical method

An alternative to the historical simulation approach for calculating risk measures, sometimes also referred to as the variance–covariance approach. This involves *assuming a model for the joint distribution of changes in market variables* and *using historical data* to estimate the model parameters. We assume that past data are a realization of this distribution, a random sample, so we can use them to estimate the distribution. The model is ideal when the change in a value of a portfolio is linearly dependent on changes in the values of the underlying market variables.

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)

When the portfolio includes options and other non-linear derivatives, *Monte Carlo simulation* can be used.

Delta normal method

If the loss distribution is normal with mean μ and standard deviation σ , VaR and ES can be determined analytically:

$$\text{VaR} = \mu + \sigma N^{-1}(X)$$
$$\text{ES} = \mu + \sigma \frac{e^{-Y^2/2}}{\sqrt{2\pi}(1-X)}$$

The model assumes that the change in the value of portfolio P is linearly related to changes in the risk factors (market variables):

$$\Delta P = \sum_{i=1}^n \delta_i \Delta x_i$$

Delta in the formula is the small change in the portfolio value for a small change in the risk factor. To calculate VaR, we only need the mean and SD of the portfolio changes, ΔP . It uses the variance-covariance matrix to calculate the risk of a portfolio. After having the volatility, it uses the above presented formulas to calculate the risk measures.

The model-building approach is ideally suited to a portfolio consisting of long and short positions in stocks, bonds, commodities, and other products. It is based on Harry Markowitz's portfolio theory. The mean and standard deviation of the value of a portfolio can be calculated from the mean and standard deviation of the returns on the underlying products and the correlations between those returns.

Advantage:

- Computationally *much faster* than historical simulation

Disadvantage:

- Daily changes in the risk factors are assumed to have multivariate normal distribution which is often not the case

The model-building approach is most often used for investment portfolios. (It is, after all, closely related to the popular Markowitz mean-variance method of portfolio analysis.) It is less commonly used for calculating the VaR for the trading operations of a financial institution. This

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)

is because financial institutions like to maintain their deltas with respect to market variables close to zero.

Monte Carlo simulation

A numerical method that produces a probability distribution by generating many random numbers from a known distribution.

Capital Requirement

1) 1996 Amendment

- Introduced **capital for market risk** as well:
 - Standardized approach (simpler banks): regulators specified how to compute market risk; capital assigned separately to each type of market exposure (debt securities, equity securities, FX risk, commodity risk and option)
 - Internal Model Approach (more sophisticated banks): calculating VaR (*10-day 99% confidence level*) and converting it to capital requirement
- Required **VaR back-testing**: calculating the 1-day 99% VaR over the previous 250 days and counting exceptions to see whether VaR is underestimated.
- The total capital required became the sum of the 8% capital from credit risk and the capital from market risk.

Fundamental Review of the Trading Book (FRTB)

It is a document proposing major revisions to the way regulatory capital for market risk is calculated, determining the capital in a much more complex way.

The *Basel I* calculations of market risk capital were based on a VaR calculated for a *10-day horizon with a 99% confidence level*. The problem was that they were typically based on the previous period (1-4 years) and were not back-tested. *Basel II.5* required banks to calculate a “stressed VaR” measure in addition to the current measure, which meant searching for a 250-day period where the observed movements in market variables would lead to significant financial stress for the current portfolio.

FRTB, introduced with *Basel III*, proposed the use of a *97.5% stressed ES*. In case of a normal distribution, the 99% VaR and 97.5% ES are almost the same, in non-normal distributions like

MARKET RISK MANAGEMENT AND REGULATION

(risk measurement, properties of risk measures, risk measurement of stocks, bonds, derivatives and portfolios, capital requirements for market risk, FRTB)

heavy tails, the ES is greater. The time horizon used for ES is based on the type of the risk factor's (market variable: int rate, credit spread, equity price etc.) liquidity. There is no back-testing in case of ES bc it is already a stressed measure.

Two approaches:

- Standardized approach: capital requirement is sum of a *risk charge* calculated using a risk sensitivity approach, a *default risk charge* and a *residual risk add-on*.
- Internal models approach: ES instead of VaR, risk factors are allocated to time horizons, and then liquidity adjusted ES is calculated

Calculations are carried out at *trading desk levels*, so it may happen that the for ex. a bank's foreign currency trading desk uses standardized approach while the equity trading desk is allowed to use internal approach.

Two books:

- Trading book: consists of instruments that the bank intends to trade
 - Positions held with short term (30 days) profits
- Banking book: consists of instruments that are expected to be held to maturity

The risk of *regulatory arbitrage* was mitigated by clearer explanations and definitions. Regulatory arbitrage is when banks allocate instruments to either of the two books to minimize capital.

FRTB distinguishes two types of credit risk exposure to a company:

- Credit spread risk: risk of a change in a company's credit spread
- Jump-to-default risk: risk of a company's default